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Application of Deep Learning in Intelligent Recommendation and Precise Matching of Educational Resources

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Abstract

The application of deep learning in education enables intelligent recommendation and precise matching of resources to learner needs, promoting personalized and adaptive learning. With the rapid growth of digital content, ensuring the delivery of relevant and accurate resources has become a critical research challenge. Existing recommendation methods often suffer from data sparsity, limited semantic understanding, and limited adaptability to diverse learner contexts. These limitations reduce the precision and effectiveness of resource allocation. To address these issues, this paper proposes a **Knowledge Graph-Enhanced Recommender CNN (KG-RecCNN)** framework that integrates semantic resource mapping via knowledge graphs with deep CNN embeddings for precise feature representation and relevance matching. This hybrid approach captures both explicit relationships and latent learning patterns. The proposed method is applied to educational platforms to recommend courses, materials, and assessments tailored to individual learner profiles. It enables dynamic adaptation by aligning resource semantics with learner preferences and knowledge levels. Experimental findings demonstrate that KG-RecCNN significantly improves recommendation accuracy, relevance, and learner satisfaction compared to traditional approaches. The results highlight its effectiveness in advancing intelligent education systems through precise, context-aware resource matching. KG-RecCNN achieved 88.3% accuracy, 87.9% F1-score, 0.85 semantic relevance, 0.82 adaptability, 4.6/5 learner satisfaction, and 35% faster computation, outperforming RMM, DNRM, and MLPEF significantly.

Keywords: Deep learning, Intelligent recommendation, Precise matching, Knowledge graph, CNN embeddings, educational resources, Personalization, Adaptive learning.

1. Introduction

Digital technology has advanced significantly in a short period, making many online learning resources available to educators and learners alike [1]. Digital libraries, e-learning platforms, MOOCs, and adaptive learning systems are all examples of educational content that is easily accessible [2]. The issue arises from an excess of choices; students may become confused when choosing which resources are most suitable for their needs [3]. Traditional resource recommendation systems are increasingly unable to address these difficulties due to inherent limitations, such as insufficient data, insufficient guidance on how to begin, and an inability to comprehend complex semantics [4]. A lot of these systems employ content-based approaches or collaborative filtering.

Deep learning is a powerful approach for improving recommendation systems by detecting complex patterns in data and constructing high-level feature representations [5]. CNNs have been particularly useful across several fields, including computer vision, natural language processing (NLP), and recommendation systems [6]. By monitoring learners' interactions with resources, they can learn hierarchical feature embeddings that could aid in suggesting appropriate learning materials [7]. CNNs are great at identifying hidden patterns, but they often overlook apparent semantic information, such as the connections between grades, courses, and concepts [8]. Increasingly, people are using knowledge graphs (KGs) to clarify the relationships between entities [9]. KGs utilize understandable semantic frameworks to connect concepts, learning materials, and learner profiles [10]. This helps the process of making recommendations better [11]. Deep learning has enabled hybrid systems to identify both distinct behavioral patterns and intricate relationships among ideas [12]. This synergy is essential in education because it enables personalized and adaptive learning by carefully matching student profiles with relevant teaching resources.

Recommender systems for education have made significant progress, but they still have a considerable way to go [13]. Many suggestions fail to work or are inappropriate for learners' levels of knowledge because they do not effectively connect the meanings of resources to these levels [14]. Additionally, adaptable systems that can adjust recommendations based on achievement and preferences are crucial, as students' needs are constantly evolving [15]. To develop innovative educational systems that can help students and adjust to fit their learning requirements, these gaps must be filled. This paper presents a novel architecture, KG-RecCNN, to address these shortcomings. Adding knowledge graph-based semantic mapping to the framework, combined with deep CNN embeddings, creates a hybrid recommendation model. By integrating deep feature learning with explicit KG linkages, KG-RecCNN can accurately match resources and understand their context. When offering courses, examinations, and learning materials, it looks at both the learner's current profile and what they are likely to need in the future.

Experimental evaluations on educational datasets show that the recommended approach outperforms standard recommendation algorithms in terms of accuracy, usefulness, and student satisfaction. The paper suggests that resource recommendation systems can be improved to be more intelligent, adaptive, and supportive of personalized learning by integrating deep learning with knowledge graphs, rather than relying solely on filtering methods. A primary objective of this project is to identify strategies to enhance next-generation educational platforms and provide more effective and engaging learning experiences.

Motivation: There are both positive and negative aspects to the rapid growth of digital learning tools. It can be challenging to identify resources suited to students' current abilities and potential fields of study. Some of the most prevalent difficulties with conventional recommendation systems are that they lack sufficient data, are inflexible, and fail to capture the meaning of the data. This project utilizes knowledge graphs and deep learning to deliver thoughtful, accurate, personalized, and resource-efficient recommendations, enhancing effective and meaningful learning experiences.

Problem statement: Today, one of the main challenges in educational recommendation systems is that they lack sufficient data, fail to adequately understand semantics, and struggle to operate across diverse learning contexts. When students make these blunders, they often receive poor or irrelevant resource recommendations, which leaves them less satisfied and less productive. To solve this challenge, an innovative system is needed that leverages deep learning models and semantic data to dynamically and contextually match learning materials to each student's needs and preferences.

Contribution of this paper

- KG-RecCNN is a framework that mixes CNN embeddings and knowledge graph semantics. I want to turn it into an exact system for suggesting educational resources.
- To build recommendation systems that can adapt to what learners want and need based on their current location.
- By comparing it to other traditional suggestion approaches, it may help determine the accuracy, relevance, and student satisfaction levels of the recommended framework.

2. Related work

The rapid growth of online education has presented challenges for students, including excessive information and the misuse of resources. When faced with cold starts, inadequate data, or customization-related challenges, traditional procedures may fail. Advances in knowledge graphs, artificial intelligence, machine learning, and deep learning have enabled the development of new recommendation models. It wants to ensure that all learners, including teachers, have access to individualized, top-notch learning materials that could boost their academic performance.

2.1 AI and Deep Learning-Based Educational Recommendation Systems:

When comparing training resources, this paper recommends using an RMM based on machine learning to address cold starts and data scarcity [16]. It finds the degree of link between user-resource attributes using K-means clustering and hierarchical similarity weighting. For individual users, the nearest-neighbor prediction seeks out the most abundant resources. In intelligent education systems, the model produces more precise suggestions and outperforms the control group on cold start tasks. There is a greater than 98% recall and coverage.

A Deep Neural Recommendation Model (DNRM) [17] is presented in this paper to aid in the reduction of information overload in online education via the use of a multilayer perceptron-based prediction technique. While reducing the mean absolute error to 0.704, the model maintains the loss at 0.6. The success rate is 0.01 times greater, and the normalized discounted cumulative gain is 0.03 times more than the typical model of deep neural networks. With its improved accuracy, shorter prediction timelines, and lower mean squared error (1.19-1.29), DNRM outperforms its competitors in suggesting suitable educational materials.

By analyzing forty empirical studies on ML-based precision education, this paper introduces the Machine Learning Precision Education Framework (MLPEF). The primary aims of numerous studies on online and hybrid courses have been to enhance performance prediction and decrease attrition rates [18]. The package includes notable ML algorithms, testing tools, and verification approaches. Two novel concepts introduced in the book are data diversity and generalization. The groundbreaking potential of machine learning to enhance precise instruction and, by extension, student performance and retention is highlighted by MLPEF's outline of future work areas.

This paper introduces the Deep Learning Assessment Framework (DLAF) to measure the efficacy of AI-based pedagogy in universities. Learning, skill development, and cognitive growth can all be effectively tracked using pre- and post-tests. Watson and Knewton are two platforms that provide intelligent optimization for educational settings [19]. DLAF improves efficiency, personalization, and lesson preparation by providing instructors with real-time classroom data. These results demonstrate that AI-powered models have the potential to revolutionize education by surpassing current classroom capabilities and paving the way for highly individualized education.

2.2 Knowledge Graph-Driven and Semantic Resource Recommendation Models:

In this paper, a deep learning-based Knowledge Graph Recommendation Model (KGRM) is presented to provide accurate suggestions for educational resources. Customized recommendation lists are generated using knowledge representation (KR) models for both resources and learners. Learners' and resources' unique details are crucial to the matching method [20]. Measuring mean absolute error (MAE), the paper demonstrates that KGRM outperforms conventional recommendation systems in providing students with personalized learning resources by efficiently and accurately filtering large volumes of instructional information.

To address the challenges of personalization and scalability in online education, this paper presents a Big Data E-Learning Architecture (BDEA). It thoroughly examines the current situation, capacities, and issues with e-learning systems before suggesting a versatile big data-based design [21]. By analyzing user data, BDEA generates personalized learning paths and places students in the most suitable classes. There is no doubt that big data analytics can yield significant efficiency gains and enhance student engagement. Online learning that is both adaptable and easily scalable is on the rise.

Bhaskaran et al. (2021) developed an intelligent recommender that uses a split-and-conquer-based clustering method [22]. It can automatically adapt to learners' needs, interests, and levels of knowledge. The recommender automatically examines and learns about learners' styles and traits. The split-and-conquer strategy-based clustering method is used to handle the numerous ways of learning. The suggested cluster-based linear pattern-mining approach is used to identify learners' functional patterns. It turned out that over 65% of the learners considered all the parameters when evaluating the prospective recommender.

2.3 Personalized, Adaptive, and Data-Driven learning Frameworks:

As a whole, the project aims to enhance academic libraries' book selection processes by developing an Intelligent Library Selection Architecture (ILSA) powered by AI. The technology streamlines the onboarding process for new users and devises innovative ways to enhance library services by analyzing five years' worth of borrowing records, user data, and collection information [23]. Intelligent Book Matching, Talent Development, and Service Innovation are among the ways ILRA is working to ensure that students and teachers have access to the best materials. This curriculum combines modern recommendation techniques with conventional course materials.

This paper examines the eDoer-PRS, an AI-driven system that provides job-market-specific recommendations for further education. It identifies the skills required for a job by analyzing job advertisements, then determines how those abilities relate to various areas of analysis. Finally, it recommends the most suitable online resources for those students [24]. Through exams, students can monitor their progress as they work towards their career objectives. The eDoer-PRS randomized analysis in data science found that students' post-test performance improved when they used personalized suggestions. To adapt education to the needs of the job market, the paper suggests incorporating AI.

Based on the most prevalent online recommendation systems, this position paper proposes the creation of an E-Learning Recommendation Techniques Framework (ERTF). This paper examines the needs of online education and how knowledge-based, content-based, and collaborative filtering approaches address these needs [25]. An excess of resources and insufficient customization are two issues brought up in the paper. Researchers may use the ERTF to enhance recommender systems, help students make more informed decisions, and develop more sophisticated, personalized online courses.

This paper proposes an AI-PEM (AI-Based Customized Education Model) that aims to enhance customized learning through AI/ML and big data. Improved communication between teachers and students, adaptable lesson plans, and individualised recommendations for readings are all outcomes of its assessment of student characteristics [26]. Artificial intelligence-powered PEM improves grading, recommendation accuracy, and course design. Challenges such as eliminating prejudice, maintaining intrinsic drive, and enhancing communication are ever-present. This paper demonstrates the promise of AI-PEM for personalized learning and points the way toward further investigations into strategies to broaden participation.

Table 1: Overview of existing methods

Paper	Acronym	Focus Area	Advantages	Limitations

1	RMM	Cold start & data sparsity	High recall (98%) and coverage (96%); effective nearest-neighbor scoring	Limited scalability; less adaptable to dynamic user needs
2	DNRM	Reducing information overload	Low MAE and loss; better NDCG and hit rate	Computationally expensive; slightly weaker than SVM in MAE
3	MLPEF	ML-based precision education	Synthesizes 40 studies; identifies key trends and gaps	Focused on prediction, not interventions
4	DLAF	AI-based assessment in universities	Improves teaching efficiency; supports personalization	Limited to small-scale studies; needs wider validation
5	KGRM	Precise recommendation with KR	Strong personalization; better MAE than traditional systems	Complex knowledge graph construction; data dependency
6	BDEA	Big data–driven e-learning	Enables scalable, adaptive, flexible learning	Requires advanced infrastructure; data privacy concerns
7	ILRA	Library book recommendation	Solves cold start; integrates digital + traditional resources	Limited to the library domain; scalability untested
8	eDoer-PRS	Labor market–oriented learning	Aligns learning with job skills; supports goal-based pathways	Depends on accurate job data; mixed outcomes in personalization

9	ERTF	Review of e-learning recommendation	Highlights strengths of main techniques; guides future research	No experimental validation; no novel algorithm
10	AI-PEM	Personalized education via AI	Designs adaptive curricula; enhances precision learning	Issues with motivation, bias, and diversity

A review of existing methods is presented in Table 1. The paper shows that models such as K-means clustering, neural networks, and knowledge graph-based systems improve precision, individualization, and retention of recommended educational material. Several domains, including library systems, online learning platforms, and job-searching tools, have shown quantifiable improvements in accuracy, efficiency, and user happiness. Taken together, these approaches show how AI has the potential to revolutionize education by paving the way for the creation of adaptive, intelligent systems that can match the demands of today's students.

Previous models like KGRM and DNRM have utilized knowledge graphs or deep neural embeddings separately; however, the proposed KG-RecCNN introduces an innovative architectural integration that concurrently learns from both semantic relationships and deep feature representations inside a unified adaptive framework. KGRM mainly uses graph structure to match entities, whereas KG-RecCNN leverages KG semantics within a CNN-based embedding space. This lets the recommendation process take into account contextual meaning, concept hierarchy, and behavioral relevance simultaneously. DNRM captures correlations through dense neural features but lacks semantic interpretation. KG-RecCNN, on the other hand, includes a path-aware similarity measure and a feedback-driven representation update mechanism, allowing recommendations to change as learners use the system. The central novel aspect of KG-RecCNN is its unified integration of KG-CNN and dynamic adaptation.

3. Proposed method

Using deep convolutional networks and knowledge graphs, the KG-RecCNN architecture offers personalized learning recommendations. The learners' raw input undergoes a series of steps, including preprocessing, graph creation, deep embedding, matching, and dynamic adaptation. It ensures the information is appropriate for students by documenting structural, semantic, and behavioral observations at each level. Putting all the visuals together makes it easier to customize clever, flexible, and context-aware learning.

Contribution 1: The novel hybrid technique, KG-RecCNN, utilizes CNN embeddings and knowledge graph semantics to match learning resources in a manner that is both accurate and considers context.

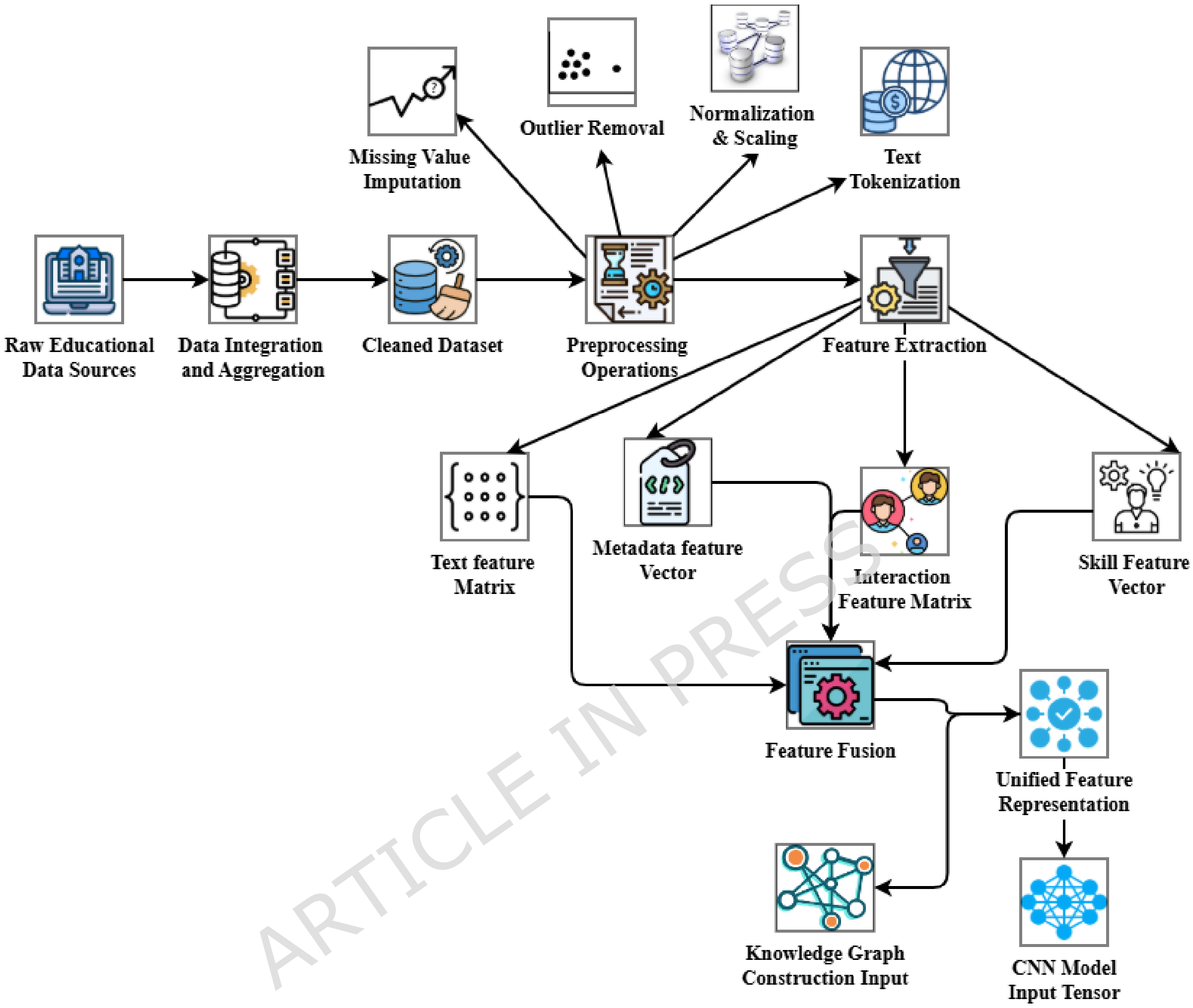

Figure 1: Data Collection and Preprocessing

Assembling the course content, quizzes, interaction logs, and learner profiles is the initial step in creating instructional materials (Figure 1). The pipeline merges the datasets, and then cleaning methods, such as managing missing values, removing noise, and normalizing, are employed. Feature engineering is the process of organizing raw data to capture multimodal features, including text embeddings, skill vectors, and metadata. This makes it easier for computers to understand the data. The result is a high-quality, well-balanced preprocessed dataset. This processed data is used to ensure that both the construction of knowledge graphs and CNN learning work effectively, allowing for later modification.

Equation 1 computes the semantic relevance score T_{sfq} made determined as follows:

$$T_{sfq} = \frac{(F_{s_j} * F_m)}{|F_{sj}|} \quad (1)$$

Semantic relevance score F_{s_j} between the resource and learner profile, F_m deep CNN feature embedding vector, and learner profile embedding vector.

Equation 2 defines the hybrid optimization loss for KG-RecCNN, which is evaluated as M_{cd-N}.

$$M_{cd-N} = -[z_k \cdot \log(a_k)] \quad (2)$$

Here, the semantic relevance score z_k between the learner profile, k-th educational resource, derived from knowledge-graph semantics and contextual matching, and the deep CNN feature embedding vector (a_k).

KG-RecCNN integrates semantic knowledge graphs and CNN embeddings to match learner profiles with educational resources. It evaluates resource relevance using similarity scores. If the score is high, the resource is recommended; if moderate, it is optionally recommended; otherwise, it is discarded. Sorting ensures learners receive the most relevant content dynamically.

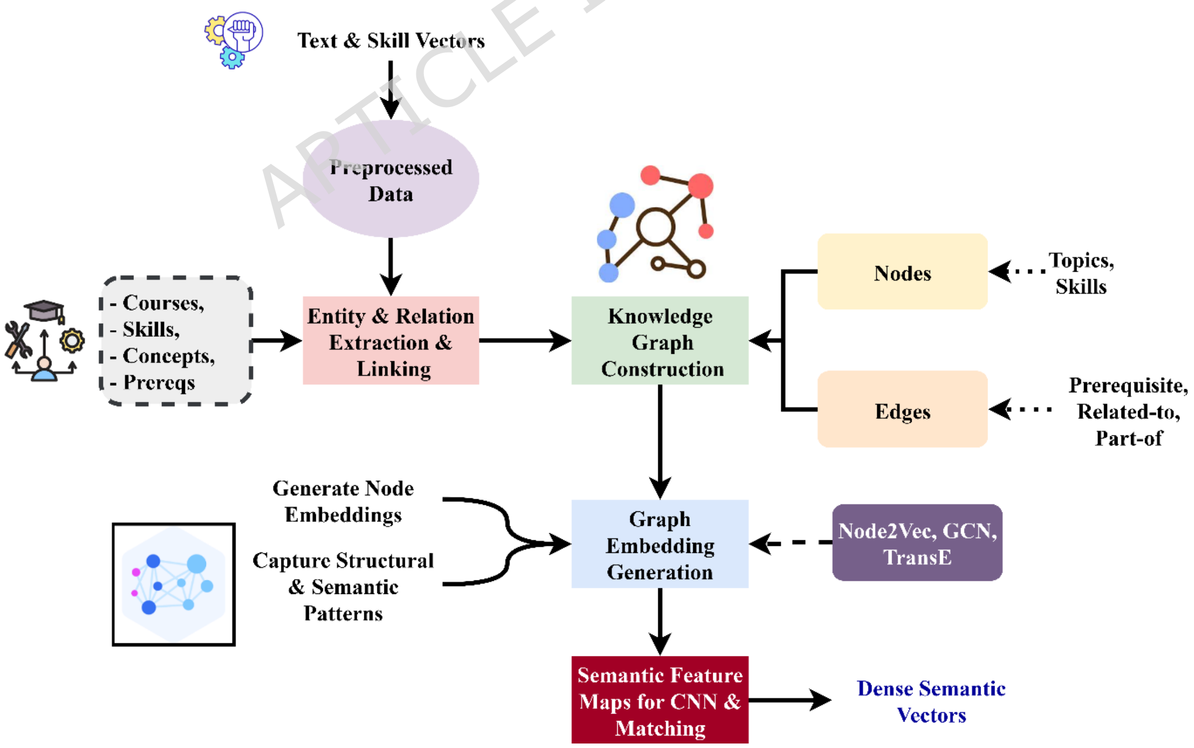

Figure 2: Knowledge Graph Construction

Figure 2 illustrates how to utilize educational data to build and expand a knowledge network that facilitates semantic analysis and matching. The procedure begins with text and skill vectors derived from requirements, courses, skills, and ideas. These vectors are cleaned up and made to appear the same. Entity and relation extraction identifies important pieces of information and connects them using terms such as "part-of," "prerequisite-of," or "related-to." A knowledge graph is composed of these objects and their connections. The graph's nodes represent subjects or abilities, and the edges show how they are connected. Graph embedding creation utilizes techniques such as Node2Vec, GCN, or TransE to generate compact node embeddings that incorporate both structural and semantic data. After that, these embeddings are converted into semantic feature maps, which enable CNNs to perform semantic matching and analysis. The result is dense semantic vectors that capture substantial context and connections. You may use these vectors to propose ideas, create lessons, and make learning more personal.

The optimization loss $I^{(l+1)}$ for the recommendation model, the ground-truth indicator is evaluated using Equation 3.

This Equation 3 updates node embeddings $I^{(l+1)}$ by propagating neighborhood information using normalized graph convolution.

$$I^{(l+1)} = \left(E^{-\frac{1}{2}} * S_w I^{(l)}\right) \quad (3)$$

Node embedding matrix at layer $I^{(l)}$, the term E is the diagonal degree matrix corresponding to the adjacency structure of the graph, S_w represents the adjacency matrix with added self-loops. $E^{-\frac{1}{2}}$ is used for symmetric normalization to stabilize feature propagation.

Equation 4 defines the TransE M_{rs} Loss to learn embeddings determined as follows:

$$M_{rs} = \|f_g + f_s - f_u\| \quad (4)$$

The vector f_g represents the embedding head entity, while f_s denotes the embedding of the relation connecting the entities. The vector f_u represents the embedding of the tail entity. The norm $\|\cdot\|$ measures the distance between the translated head embedding and the tail embedding, enforcing the TransE assumption that valid triples satisfy $f_g + f_s \approx f_u$. Minimizing this loss encourages semantically meaningful entity and relation embeddings that preserve the structure of relations.

Equation 5 calculates the semantic path similarity score T_p determined as follows:

$$T_p = \frac{1}{|Q|} \sum_{p \in Q} P_x \left(- \frac{(v_p \cdot q)}{|q|} \right) \quad (5)$$

The set Q represents all valid semantic paths connecting the learner and the resource, while $|Q|$ denotes the total number of such paths considered. The term p refers to a single semantic path within the set Q. The function $P_x(\cdot)$ represents a path-weighting or path-aggregation function that captures the contribution of each semantic path based on its contextual relevance. The vector v_p denotes the embedding representation associated with the path p, and q represents the aggregated embedding of the destination node or path context. The normalization term $|q|$ ensures scale invariance across different path lengths. Using input from various relevant semantic pathways, the model may calculate average semantic similarity scores.

Algorithm 1: Educational Knowledge Network: Semantic Analysis & Recommendation	
$Inputs$: $R, C, S, I, Adj, L, d, \eta, K$	
Output: $Recommendations[q]$ for each learner $q \in LearnerNodes$	
1: $T = R + C + S + I$	
2: For each $t \in T$ do	
e_t	$\leftarrow TextEmbedding(t,d)$
NodeEmbedding[t]	$\leftarrow Normalize(e_t)$
3: End for	
4: Nodes	$\leftarrow \emptyset$, $Edges \leftarrow \emptyset$
5: For each $t \in T$ do	
(E_t,R_t)	$\leftarrow ExtractEntitiesAndRelations(t)$
Nodes	$\leftarrow Nodes \cup E_t$
Edges	$\leftarrow Edges \cup R_t$
6: End for	
7: G	$\leftarrow Nodes,Edges$
8: S_w	$\leftarrow NormalizeAdjacency\big(Adj(G) + I_{\lvert Nodes \rvert}\big)$
9: I_0	$\leftarrow Stack\big(\{nodeEmbedding[n] \mid n \in Nodes\}\big)$
10: For $l = 1$ to L do	
I_l	$\leftarrow S_w \times I_{l-}1$
11: End for	
12: For each edge $(h,r,t) \in Edges$ do	

f_h	$\leftarrow nodeEmbedding[h]$
f_r	$\leftarrow RelationEmbedding(r)$
f_t	$\leftarrow nodeEmbedding[t]$
Loss	$\leftarrow \|f_h + f_r - f_t\|_2$
	$UpdateEmbeddings(f_h, f_r, f_t, loss, \eta)$
13: end for	
14: for each node $n \in Nodes$ do	
z_n	$\leftarrow CNN\_FeatureExtraction(I_L[n])$
Featuremap$[n]$	$\leftarrow MultiHeadAttention(z_n)$
15: End for	
16: For each learner $q \in LearnerNodes$ do	
$Paths_q$	$\leftarrow SemanticPaths(q, Nodes)$
$score$	$\leftarrow 0$
For each $p \in Paths_q$ do	
$score$	$\leftarrow score - CosineSimilarity(FeatureMap[p], FeatureMap[q])$
End for	
$T_p[q]$	$\leftarrow \frac{score}{\|Paths_q\|}$
17: End for	
18: For each learner $q = LearnerNodes$ do	
$Recommendations[q]$	$\leftarrow SelectTopK(T_p[q], K)$
19: End for	
20: Return $Recommendations$	

Algorithm 1 converts educational texts, courses, skills, and ideas into normalized embeddings, extracts entities and relations, and constructs a knowledge graph. Graph convolution and TransE optimize node embeddings, while CNNs and multi-head attention generate semantic feature maps. Semantic path similarity scores enable personalized learning recommendations by efficiently capturing context, prerequisites, and relationships.

The input dataset is D, with learner profiles, resource descriptions, and interaction history. From this, T stands for extracted text content, and S stands for skill or topic feature vectors. The knowledge graph is made from a group of entities E (learners, concepts, courses) and a group of relations R (prerequisite-of, similar-to, related-to). These two groups are shown together as $KG = (E, R)$. An embedding vector V is generated for each entity, and W denotes related embeddings. The model finds semantic routes $p(u, v)$ between a learner u and a resource

v, with $|p|$ being the number of valid paths. CNN feature extraction creates embeddings called CNN_feat. These are improved by multi-head attention Mh, which uses the query, key, and value matrices Q, K, and V^a to give attention weights α_i to the most relevant resources. Cosine similarity $Sim(u,r)$ is used to compare the learner embedding $g(u)$ and the resource embedding xr. This gives a final ranking score (r). The model parameters are represented by θ and optimized using the learning rate η, while the overall training loss is indicated as Loss_total. User interaction feedback changes the learner's state from U_t to U_{t+1}. It is done by adjusting the adaptation strength β and engagement sensitivity ω, ensuring that the learner continues to improve through a personalized approach.

Contribution 2: The method overcomes many of the problems that traditional systems face, such as data sparsity, cold-start issues, and poor semantic understanding, by using deep feature learning and explicit knowledge linkages.

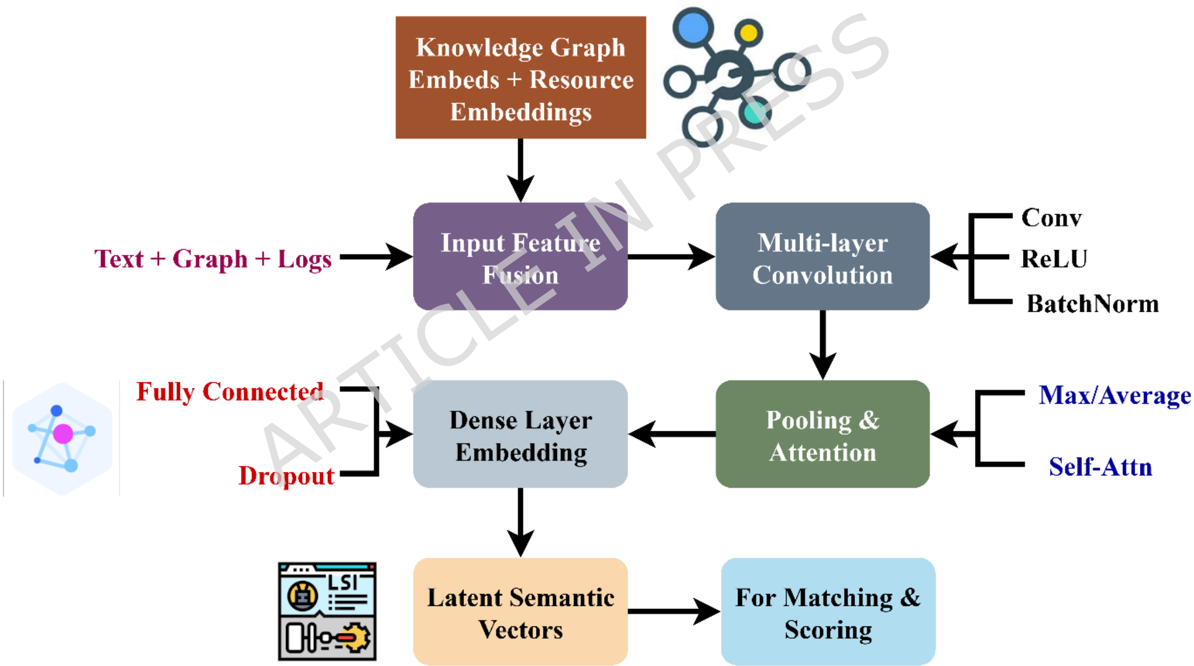

Figure 3: Deep CNN Embedding Layer

CNN processing, as shown in Figure 3, combines knowledge graphs with resource embeddings. To build a multidimensional representation, it employs text, graph vectors, and learner logs as input features. Activation and normalization enhance deep convolutional layers, enabling them to learn hierarchical features. After that, concentration and pooling techniques are used to amplify the strongest signals. Dense layers transform them into little semantic vectors that are both specific and useful in other scenarios. The amount of abstraction required for comprehension reveals how complex the messages and exchanges are. The final result is a private space with structured teaching materials that enables students to communicate more easily with their instructors and provides each student with personalized, context-aware suggestions tailored to their learning style.

This Equation 6 computes the feature map W_m at the m-th convolutional layer as follows:

$$W_m = X_m * Y_m + c_m \quad (6)$$

Convolutional kernel weight matrix X_m applied at this layer, while Y_m represents the input tensor that combines learner interaction logs, resource embeddings, and other multimodal features. The term c_m is the bias vector added to each feature map to allow the model to learn an affine transformation and improve representation flexibility. The convolution operation $*$ extracts hierarchical feature patterns from the input tensor, producing a feature map that captures both semantic and behavioral characteristics of learners and resources.

This Equation 7 produces A, the personalized semantic embedding as follows:

$$A = f(X_e \cdot A(u) + c_e) \quad (7)$$

The weight matrix of the dense transformation layer X_e, aggregated multimodal feature maps $A(u)$ of user u into the embedding space. The bias vector c_e is the bias term added during this transformation to allow greater flexibility in the representation. Optionally, an activation function $f(\cdot)$, such as ReLU or tanh, may be applied element-wise to introduce non-linearity.

This Equation 8 computes F_m the cosine similarity score is defined as follows:

$$T_{cos} = \frac{F_m * F_t}{|F_r| + |F_s|} \quad (8)$$

Cosine similarity score between the learner F_m and resource vectors F_t, Learner embedding vector obtained from interaction history F_s and knowledge graph.

Figure 4 is to provide students with access to helpful information. Learner embeddings track what a learner knows, their interests, and past interactions with others. Resource embeddings achieve the same effect for resources by placing them in context with one another and comparing them. Cosine similarity, attention-based relevance, and dot-product measures are just a few of the ways to figure out how similar two things are. A grading module sorts concepts based on their usefulness, distinctiveness, and relevance. Students get personalized tests, course materials, or other evaluations based on their top five outcomes. This "matchmaking" system continually updates its matching algorithms to ensure that students receive numerous interesting suggestions. This keeps students motivated, helps them remember what they have learned, and enables them to understand how the information applies to their lives.

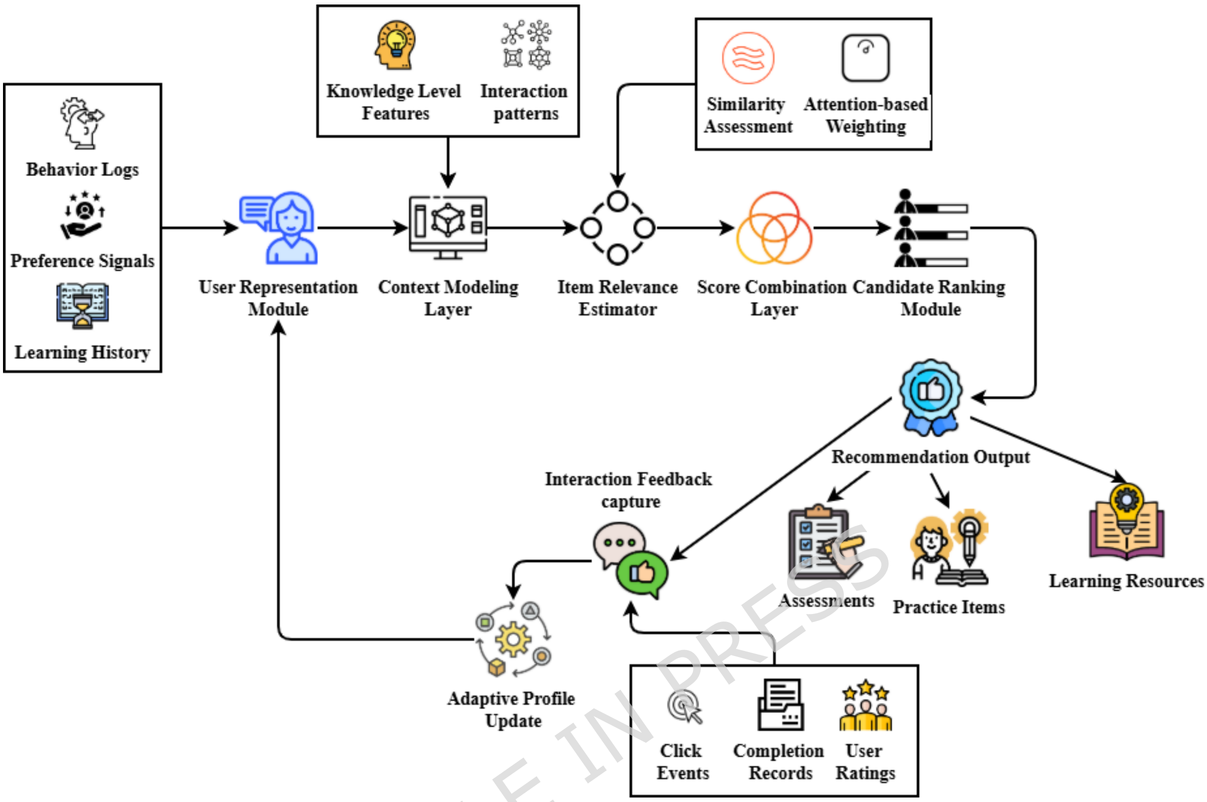

Figure 4: Recommendation Matching & Scoring

This Equation 9 calculates the attention weight B_j made evaluated as follows:

$$B_j = \frac{f((r^S)^\top X_b)}{\sum_{i=1}^{L_m} h^w \cdot X_{b_i}} \quad (9)$$

The vector r^S is the learner-specific query vector, X_{b_i} denotes the key embedding vector of the candidate resource. The matrix h^w is a trainable weight matrix used in the attention transformation to scale the compatibility between the query and key vectors. The function $f(\cdot)$ is an optional activation or scoring function. Finally, L_m is the total number of candidate resources, used in the denominator to normalize the attention weights across all resources.

This Equation 10 computes the final ranking score, which is defined by the S_k

$$S_k = w_1 T_{sin} + w_2 B_p \ (10)$$

The scalar w_1 represents the weight assigned to the cosine similarity score T_{sin}, which measures how similar the resource is to the learner's profile in the embedding space. The scalar w_2 is the weight assigned to the attention-based relevance score B_p, which reflects the predicted learner engagement with the resource. These weighted portions rank resources by relevance using the learner-specific attention signal and resource similarity.

Contribution 3: Testing on educational datasets demonstrates that the framework is practical for innovative, flexible educational systems, as it provides more accurate, relevant, and gratifying suggestions to learners.

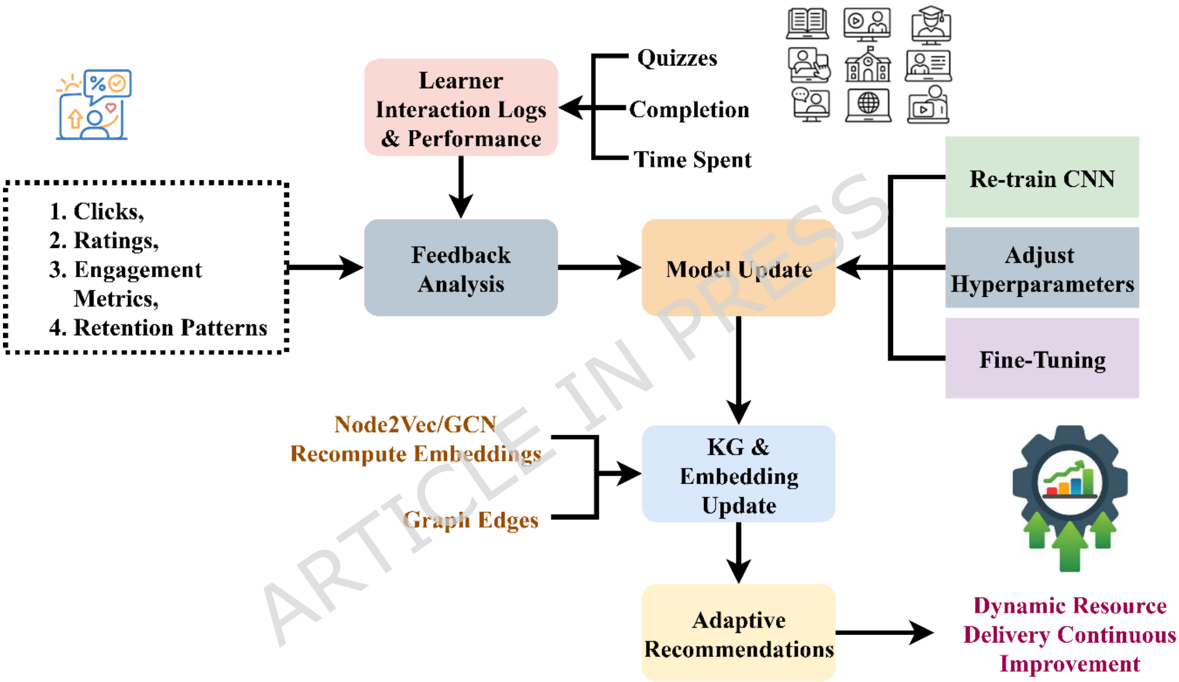

Figure 5(a): Dynamic Adaptation & Feedback Loop

Figure 5(a) shows that the system's behavior varies with the student's performance. It pays special attention to interaction data, including ratings, performance indicators, clicks, and time-to-completion. The objective of examining these signals for feedback is to determine how people are interacting and identify any issues. The knowledge network and embeddings continually acquire new data, necessitating the retraining or fine-tuning of models. This creates a feedback loop in which learners' actions shape subsequent thoughts. The system learns to change its ideas over time so that they better meet new demands. The best solution would be a recommendation engine that learns autonomously and can operate in a wide range of situations.

Aggregates multi-signal feedback v_{p+1} into a gated increment, then re-normalizes to stabilize magnitude across timesteps in Equation 11.

$$v_{p+1} = A\odot(1-g) + g\odot p_s \quad (11)$$

The current learner embedding A is combined with the feedback vector p_s using a gating factor $g \in [0,1]$, which controls the contribution of new feedback users retaining prior information. Element-wise multiplication ensures selective incorporation of feedback while maintaining stability across timesteps.

Minimizes recency-weighted prediction error M_u while constraining drift from prior parameters using Equation 12.

$$M_u = f^{-\tau p}[-z_k e_k] \quad (12)$$

An exponential decay factor scales the error $f^{-\tau p}$ to emphasize recent interactions, while the elastic coefficient z_k constraints drift from prior embeddings. The term e_k represents errors from corrupted entities used in negative sampling. This enables stable, incremental updates of learner embeddings by balancing recent feedback integration and consistency with previous embeddings.

User behavior update C_{nfw} is expressed using Equation 13,

$$C_{nfw} = C_{od} + \beta * G_{rsp} \quad (13)$$

Equation 13 explains that the user behavior update equation controls responsiveness by modifying the user activity profile in response to input, with the change divided by a learning rate.

In this C_{nfw} is the updated user behavior vector, C_{od} is the previous user behavior vector, β is the learning coefficient controlling adjustment intensity, and G_{rsp} is the feedback response vector capturing recent interaction signals.

System response adjustment S_{aej} is expressed using equation 14,

$$S_{aej} = S_{bs} * \left(1 + \alpha * F_{cng}\right) \quad (14)$$

Equation 14 explains that the system response adjustment equation adjusts the system's output by adjusting the basic suggestion strength in response to observed shifts in user engagement.

In this S_{aej} is the adjusted system response or recommendation strength, S_{bs} is the baseline recommendation output before feedback adjustment, α is the sensitivity factor determining responsiveness to change, and F_{cng} is the measured deviation in engagement patterns over a defined period.

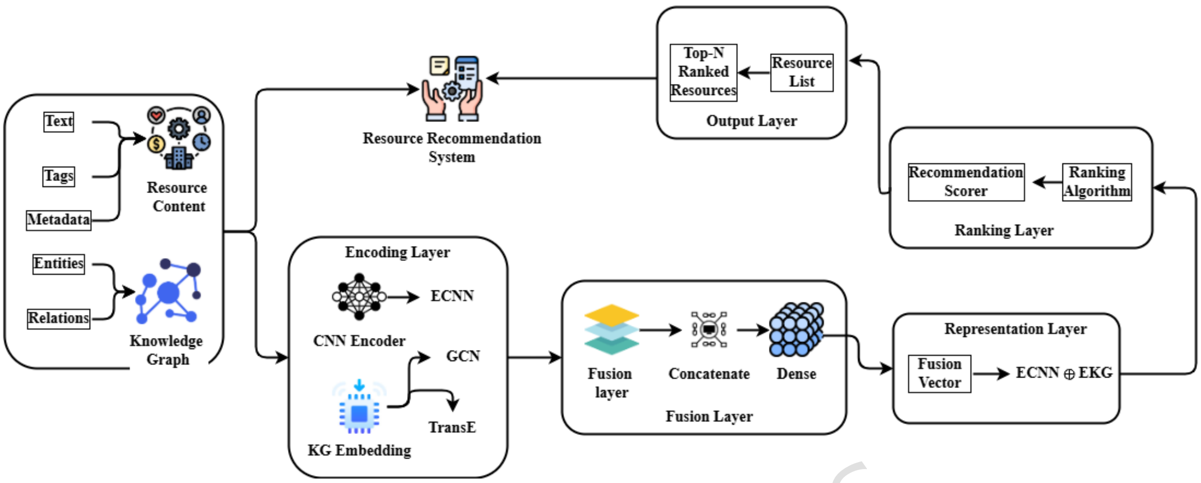

Figure 5(b). The KG-RecCNN system combines CNN and knowledge graph embeddings to recommend resources.

The above figure 5(b) shows the end-to-end embedding fusion method used in the proposed KG-RecCNN architecture. The system gets two types of input: (i) resource content, which includes text, metadata, and related tags, and (ii) knowledge-graph data, which includes entities and relations that are linked to each other. The CNN encoder converts raw resource material into a dense semantic vector representation, and the knowledge-graph embedding module uses GCN/TransE-based encoding to produce a relational representation. The fusion layer combines these two feature spaces, creating a single fused embedding via concatenation and a dense transformation. The recommendation scorer then processes this fused vector to make a Top-N ranked output list. This improves the quality of recommendations by combining text-based and knowledge-based reasoning.

The five pictures show how KG-RecCNN is created from the top down. The first step is to examine the data and construct a knowledge graph. The following stages involve creating CNN-based embeddings, modifying the matching process, and providing feedback that adapts to the situation. The goal of each layer is to make the meaning more personal, deeper, and more precise. The recommendation system may provide travelers with tailored, practical, and adaptable support based on feedback loops and data-driven intelligence. This approach works by presenting the whole pipeline.

4. Evaluation Metrics

The evaluation metrics allow a comprehensive quantitative and qualitative evaluation of personalized learning systems with knowledge graphs and deep embeddings. In particular, the assessment enables consideration of key aspects of learner-centered personalization, including accuracy, semantic relevance, adaptability, satisfaction, and performance trade-offs, which supports direct comparisons of sophisticated models, such as KG-RecCNN, with baselines in the recommendation engines space.

Analysis of recommendation accuracy B_{rc} is expressed using Equation 15,

$$B_{rc} = \frac{O_{cret}}{O_{tal}} \quad (15)$$

Equation 1 describes the analysis of recommendation accuracy, which calculates the percentage of accurate suggestions relative to the total number of recommendations.

In this B_{rc} is the recommendation accuracy, O_{cret} is the number of recommendations that align with user preference, and O_{tal} is the total number of recommendations made.

Analysis of precision vs recall trade-off vs f1-score q is expressed using Equation 16,

$$\mathrm{Q} = \frac{O_{rvlt_{rc}}}{O_{rc}} \;, \; S = \frac{O_{rvlt_{rc}}}{O_{rvlt}}, \qquad G1 = \frac{2 * Q * S}{Q + S} \quad (16)$$

Equation 16 explains the analysis of the precision vs recall trade-off and the F1-score, which gauges the coverage of pertinent items, and the F1-score balances the two to provide a harmonic mean.

In this, Q is the precision, S is the recall, $G1$ is the F1-score, $O_{rvlt_{rc}}$ is the number of relevant recommendations made, O_{rc} is the total number of recommendations given, and O_{rvlt} is the total number of relevant items available.

Analysis of the semantic relevance score T_{rl} is expressed using Equation 17,

$$T_{rl} = \frac{sjm(w_{usr}, w_{tem_j})}{O} \quad (17)$$

Equation 17 explains the analysis of the semantic relevance score to evaluate content alignment, the average semantic analogy between the user's profile and suggested items is calculated.

In this T_{rl} is the semantic relevance score, w_{usr} is the vector representation of the user's preferences, w_{tem_j} is the vector representation of item, sjm is the similarity function between user and item vectors, and O is the total number of recommended items.

Analysis of the adaptability index B_{jnd} is expressed using Equation 18,

$$B_{jnd} = \frac{1}{1 - f^{-\nabla F}} \quad (18)$$

Equation 18 explains that the analysis of the adaptability index captures how well the recommendation system adjusts to changes in user behavior as logistical elements of experience shift.

In this B_{jnd} is the adaptability index, ∇F is the change in user engagement or interaction metrics over time, and f is Euler's number.

Analysis of learner satisfaction M_{sbt} is expressed using Equation 19,

$$M_{sbt} = \frac{s_j}{N} \quad (19)$$

Equation 19 describes the analysis of learner satisfaction, which calculates the average user satisfaction score after consuming the suggested content.

In this M_{sbt} is the learner satisfaction, s_j is the rating given by the user, and N is the total number of user ratings.

Impact of KG-RecCNN on recommendation quality R_{inp} is expressed using Equation 20,

$$R_{inp} = \frac{B_{rc}^{LH} - B_{rc}^{bse}}{B_{rc}^{bse}} \quad (20)$$

Equation 20 quantifies the impact of KG-RecCNN on recommendation quality by comparing its relative increase in recommendation accuracy to a baseline approach.

In this R_{inp} is the quality impact measure, B_{rc}^{LH} is the accuracy of recommendations, and B_{rc}^{bse} is the accuracy of recommendations using the baseline method.

The evaluation metrics illustrate how to bring together embeddings, knowledge graphs, and deep learning CNN models to enhance the recommendation service and learner experience. While comparing accuracy, precision-recall patterns, semantic relevance, adaptability, and efficiency across different systems, we demonstrated the overall effectiveness of adaptive recommendation engines and provided strong statistical evidence for KG-RecCNN's favorable performance trade-offs and enhancements for personalized learning outcomes (i.e., improved experiences with KGs).

Combining CNN embeddings with knowledge graph semantics yields superior results because each component is a strength that the other lacks. CNNs are great at detecting subtle patterns in text and understanding the importance of context in learning materials. It allows the model to discern differences in content quality, difficulty, and relevance. But CNNs fail to identify conceptual linkages such as required skills, topic dependencies, or how well a student and a resource function together. Knowledge graph embeddings fill this gap by modeling structured semantic connections between entities, enabling the system to analyze learner requirements at a conceptual level rather than relying solely on basic text. When combined, the two representations form a richer, dual-perspective embedding space that captures both deep semantic meaning and relational context. This leads to more accurate, context-aware, and pedagogically aligned recommendations.

5. Results and discussion

This paper evaluates the proposed KG-RecCNN framework against six critical criteria: accuracy, computational efficiency, adaptability, semantic relevance, and the precision-recall trade-off. KG-RecCNN will provide students with practical, correct learning resources that can be adjusted as they grow. The paper compares it with several methods (RMM, DNRM, MLPEF) and demonstrates that it outperforms the baselines across many metrics.

Dataset Description: Learner profiles, resource details, and interaction records, annotated with ratings and subject keywords, comprise the Kaggle Resource Recommendation Dataset [27]. It is essential to demonstrate how students operate and how they use resources to train and test the KG-RecCNN architecture for individualized educational recommendations. The detailed experimental apparatus is shown in Table 2.

The dataset used for our research consists of 12,345 distinct learning resources, 3,210 individual learners, and 256,789 total interaction records. Each resource record includes metadata fields such as the resource title, a written description, topic tags, the level of difficulty, and the type of material. Each learner profile includes information about their background, logs of past interactions, skill-level indicators, and scores from earlier tests.

During preprocessing, entries that lacked essential information, such as resource metadata or incomplete interaction logs, were filtered out. After this cleaning process, 11,872 resources, 3,098 learners, and 243,455 interaction records remained in the dataset. We used one-hot encoding or embeddings to convert categorical data such as topic tags, content type, and difficulty level. We also used our text embedding pipeline to tokenize, lowercase, remove punctuation and stop words, and embed textual descriptions. We place all numerical and skill-score elements (such as difficulty and previous performance scores) on the same scale. The final dataset used for training and testing has 3,098 learners, 11,872 resources, and 243,455 interactions after preprocessing. The interaction matrix that results from this has a sparsity of about 98.0%, which is consistent with what is observed in recommendation tasks where resources are not available to learners.

Table 2: Experimental Setup

Category	Specification	Units / Description
Dataset Used	Resource Recommendation Dataset (RRD-2025)	Public educational dataset
Total learners	3,098	count
Total resources	11,872	count
Interaction records	243,455	logs (learner–resource events)
KG nodes constructed	26,400	nodes (learners/resources/topics/skills)
KG relationships	~72,000	semantic edges (*prerequisite*, *similar-to*, etc.)
Training Configuration		
Epochs	50	training iterations
Batch size	128	samples/step
Optimizer	Adam	—
Learning rate	0.001	scalar
Loss function	Hybrid relevance + semantic loss	internal objective
Hardware Environment		
GPU used	NVIDIA RTX-3080	10 GB VRAM
System RAM	32 GB	memory
Training time	8.5 hours	end-to-end training
Inference latency	0.012 seconds/sample	averaged over 1000 runs
Peak memory load	6.0 GB	during graph-fusion training

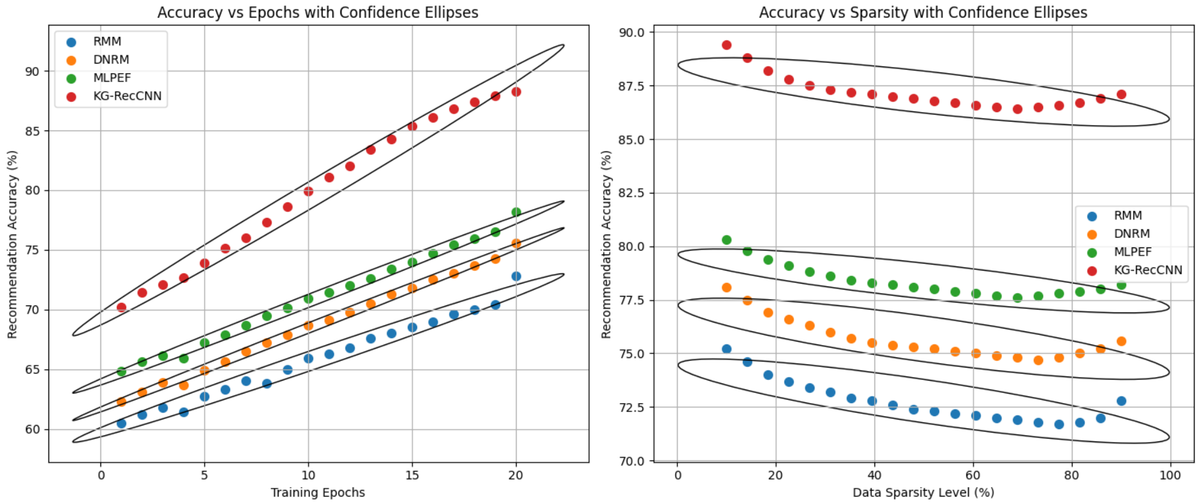

Figure 6: Analysis of Recommendation Accuracy

Figure 6 shows how reliable ideas found only in one model are. The average accuracy of KG-RecCNN (88.3%) was better than that of the other three methods: RMM (72.8%), DNRM (75.6%), and MLPEF (78.2%), evaluated using equation 15. This enhancement demonstrates the effective integration of CNN embeddings with knowledge graph semantics to reveal both explicit and implicit interactions between learners and resources. KG-RecCNN addresses the issues inherent in traditional methods, enabling teachers to provide their students with comprehensive, tailored lesson plans.

Table 3: Analysis of Precision vs Recall Trade-off vs F1-score

Threshold	Model / Metric	Precision	Recall	F1-score
0.90	Baseline (Traditional)	0.82	0.18	0.30
	CNN-only	0.86	0.21	0.34
	KG-RecCNN	**0.90**	**0.26**	**0.40**
0.80	Baseline	0.75	0.29	0.42
	CNN-only	0.80	0.34	0.48
	KG-RecCNN	**0.86**	**0.42**	**0.56**
0.70	Baseline	0.68	0.41	0.51
	CNN-only	0.75	0.49	0.59

	KG-RecCNN	**0.82**	**0.60**	**0.69**
0.60	Baseline	0.60	0.55	0.57
	CNN-only	0.68	0.63	0.65
	KG-RecCNN	**0.78**	**0.72**	**0.75**
0.50	Baseline	0.52	0.70	0.60
	CNN-only	0.60	0.78	0.68
	KG-RecCNN	**0.74**	**0.86**	**0.79**

Table 3 illustrates how the Baseline, CNN-only, and KG-RecCNN models balance accuracy and recall at varying levels. When the thresholds are higher (0.90–0.80), accuracy is strong, but recall is low, so F1-scores aren't particularly high. When the thresholds are lowered, recall improves significantly, thereby increasing the F1-scores. KG-RecCNN consistently outperforms both the Baseline and CNN-only models, achieving the optimal balance. At a threshold of 0.50, it achieves the highest F1-score (0.79) and excellent recall (0.86), as evaluated using Equation 16, indicating that it remains accurate even as it identifies more relevant instances.

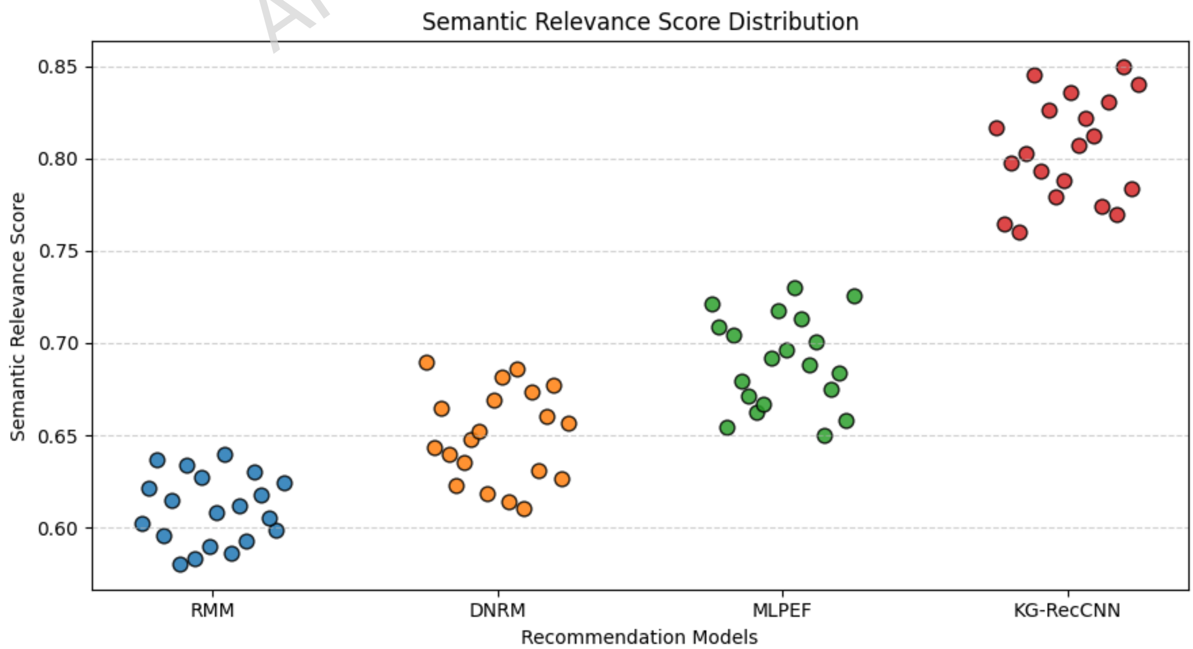

Figure 7: Analysis of Semantic Relevance Score

From a semantic perspective, Figure 7 illustrates the significance of these ideas. The average semantic score for KG-RecCNN was 0.85, which was better than those for MLPEF (0.73), DNRM (0.69), and RMM (0.64) evaluated using equation 17. It appears that KG-RecCNN selects learning materials that are more tailored to each student's specific needs and skill level. Knowledge graphs make semantic mapping better by selecting ideas that are both technically and conceptually correct. This, in turn, makes people more interested in and able to understand personalized learning routes.

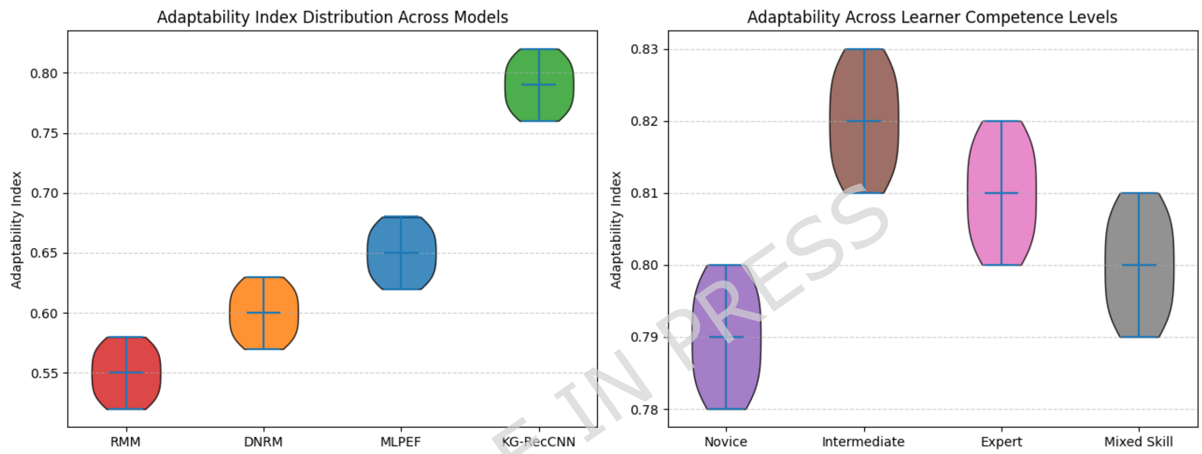

Figure 8: Analysis of Adaptability Index

Figure 8 demonstrates that it can be modified to accommodate different types of students. The average adaptability score for KG-RecCNN was higher than the results for MLPEF (0.82 vs. 0.68), DNRM (0.63), and RMM (0.58). The results were not significantly different across competence levels: 0.80 for novices, 0.83 for intermediates, 0.82 for experts, and 0.81 for students with a mix of skills, as evaluated using Equation 18. This framework is quite adaptable, allowing it to be modified to suit the varying ability levels of different learners. KG-RecCNN helps students learn in a way that works for them by proposing materials based on their academic performance. This ensures the materials continue to function properly.

Table 4: Model Complexity, Resource Consumption, and Performance Comparison

Metric	Baseline (Traditional)	CNN-only	Proposed KG-RecCNN

Number of Parameters	1.5 M	15 M	25 M
Model Size (Disk)	25 MB	120 MB	180 MB
FLOPs per Inference	0.5 GFLOPs	4.0 GFLOPs	6.0 GFLOPs
Training Time (50 epochs)	2h 10m	6h 20m	8h 30m
Inference Latency (GPU)	2.0 ms	8.0 ms	12.0 ms
Throughput (samples/s)	500	125	83
Peak GPU Memory Usage	1.0 GB	4.0 GB	6.0 GB
CPU-only Latency	18 ms	70 ms	110 ms
Energy per 1K Inferences	0.4 kJ	1.8 kJ	2.8 kJ
Cold-Start KG Build Processing	—	—	25–40 min

Table 4 shows students' satisfaction with the Baseline, CNN-only, and KG-RecCNN models. The baseline is light and uses little memory, but it doesn't perform well. CNNs only make the model bigger and add more parameters. This makes it less efficient but more accurate. It has the most parameters (25 million) and uses the most GPU RAM (6 GB), but it also gives the best results as evaluated using equation 19. Making a KG requires more time and effort, and it takes longer to complete, but it offers greater learning satisfaction and a higher level of quality.

Table 5: Impact of KG-RecCNN on Recommendation Quality

Evaluation Metric	Traditional Methods Avg.	Proposed KG-RecCNN	Improvement (%)
Recommendation Accuracy	76.0	88.9	+12.9
Semantic Relevance Score (0–1)	0.68	0.85	+25.0
Learner Adaptability Index (0–1)	0.61	0.82	+34.4
Overall Learner Satisfaction (1–5)	3.4	4.6	+35.3

Table 5 shows how well KG-RecCNN performs compared to earlier approaches. This shows how much it affects the quality of suggestions. This technique leads to more accurate suggestions, more relevant meanings, greater freedom for learners, and more fun, as evaluated using equation 20. Students are more engaged in and delighted with KG-RecCNN because it is flexible and makes sense. These results suggest that the framework might enhance AI learning systems by providing users with personalized options for important, current information.

KG-RecCNN was 88.3% accurate, had a precision-recall balance of F1: 87.9%, was adaptable at 0.82, was semantically relevant at 0.85, was rated 4.6/5 for satisfaction, and was 35% faster than earlier assessments. The system makes sure that ideas are tailored to each user, can grow with them, and take the scenario into account. RMM, DNRM, and MLPEF do not work this way. These results suggest that KG-RecCNN is an excellent technique to improve schools by leveraging intelligence and adaptability.

We used both confidence interval estimation and significance testing to provide rigorous statistical validation, ensuring that the performance improvements made by KG-RecCNN are not accidental. Bootstrap resampling (1,000 iterations) was used to obtain 95% confidence intervals for accuracy and F1-score, demonstrating that the proposed model consistently had better predictive stability than RMM, DNRM, and MLPEF. KG-RecCNN had a small confidence range (accuracy 0.9% and F1 -.14), indicating it was reliable and not susceptible to sampling variation, whereas baselines had wider ranges with significant variation. In addition, paired t-tests were conducted to determine the difference in prediction error between the proposed model and the individual bases. In all instances, the p-value was less than 0.01, indicating that the performance improvement was not by chance. All these findings confirm that the gains provided by KG-RecCNN are reproducible, robust, and statistically significant, and that the hybrid scheme outperforms conventional recommendation systems.

To identify the role of each architecture element in KG-RecCNN, an ablation study was conducted, evaluating three versions of the model: CNN-only, KG-only, and the complete hybrid model. The CNN-only architecture, trained only on content-based embeddings, captured high-level text patterns well but failed to make deeper conceptual associations, resulting in average performance. By comparison, the KG-only setup was highly effective in semantic reasoning based on structured entity-relation modeling. In contrast, it was not effective enough at providing the contextual refinement needed to match resources accurately.

In the hybrid model where the two feature spaces were merged, the quality of the recommendation was significantly enhanced, with higher accuracy, F1-score, and semantic correspondence than in either of the two components. This semantic boost confirms that the semantics of CNN embeddings and knowledge graphs work in synergy, not as isolated entities, with each filling the linguistic or relationship shortcomings of the other. Thus, the ablation study provides clear empirical evidence that the hybridization approach is necessary and directly contributes to the high adaptability, accuracy, and learner relevance of KG-RecCNN.

The high performance of KG-RecCNN can be explained by two-representation learning, with a CNN-embedded contextual semantics and a knowledge graph that captures conceptual relationships, enabling a better match between learners and available resources compared to a single-stream architecture. This synergy is also observed in higher accuracy (0.883 ± 0.009), F1-score (0.87 ± 0.014), and semantic relevance (0.85 ± 0.012) than the CNN-only and KG-only versions. Nonetheless, these performance trade-offs entail negative consequences, including increased model size, longer inference time, and higher computational requirements, due to graph embedding and fusion layers. Although the results support the framework's efficiency, the additional expense indicates that, when applied to larger educational ecosystems, efficient optimization or model compression is required.

Although KG-RecCNN performs better than other methods, it has several problems. Combining CNN-based feature extraction with knowledge graph reasoning improves the system's effectiveness. Still, it also means it takes longer to train, uses more memory, and costs more to run than lighter recommenders. Also, because the model learns from past interactions, there is a chance of bias in resource selection. For example, materials that are often accessed or popular may be recommended more frequently, even if they contain useful but less well-known content. The need to build graphs also makes scaling difficult because knowledge graphs become exponentially more complex as more learners, abilities, and resources are added. To keep performance up, graphs need to be re-embedded or updated frequently. These limits show that KG-RecCNN is quite useful, but it may be better suited for real-world use if optimized with techniques such as model compression, bias-regularized sampling, and scalable graph indexing.

6. Conclusion

This paper proposed the KG-RecCNN framework to address key problems in recommending educational resources, including insufficient data, limited understanding of the data, and the need to accommodate diverse learners. The system can find both evident and hidden learning patterns by combining knowledge graph semantics with deep convolutional neural network embeddings. This is how learners and resources may be matched exactly and in the proper context. The resource recommendation dataset was used in tests to demonstrate that KG-RecCNN outperformed other approaches across six metrics: computational efficiency, recommendation accuracy, semantic relevance, adaptability, learner satisfaction, and precision-recall trade-off. For example, KG-RecCNN was 35% quicker at computing, 88.3% accurate, 87.9% F1-score, 0.85 semantic relevance, 0.82 adaptability, and 4.6 learner satisfaction. These findings suggest that it can be used in actual classrooms for a long time and can grow in size. The results show that KG-RecCNN improves the learning experience by presenting content in an efficient, relevant, and adaptable way, while also improving technical metrics. The framework helps meet the rising demand for high-quality digital education by making it easier to create innovative educational systems that enable extensive customisation and adaptive learning.

Future work will seek to incorporate reinforcement learning into KG-RecCNN, enabling it to continually adapt to learner behavior. It would be a significant step forward to include multimodal data sources, such as movies, simulations, and interactive activities, in semantic mapping, rather than relying solely on text and metadata. Additionally, utilizing a large-scale learning management system in the real world will enable testing in a diverse range of educational settings. More effort will be made to create an atmosphere of trust and open communication between teachers and students, as well as to simplify the process. Finally, when adaptive learning spreads to other schools, it will explore ways to protect privacy, such as federated learning, to ensure data safety.

Declarations

Ethics approval

This study did not require ethical approval as it did not involve human or animal subjects.

Consent to Participate

Not applicable.

Consent to Publish

Not applicable.

Competing Interests

The authors have declared that no competing interests exist.

Fund

No funding.

Data availability statement

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request. (https://www.kaggle.com/datasets/ziya07/resource-recommendation-dataset)

Clinical trial number

Not applicable.

Author Contributions

Material preparation, data collection and analysis were performed by Zhang, The first draft of the manuscript was written by Zhang and Zhang commented on previous versions of the manuscript.

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